A Polished Weather Forecast

You are starting a job as an IT specialist in a weather station. To handle the large number of forecasts every day, the station uses the Polish logical notation[1] to store forecasts on their computer systems in a concise and unambiguous manner. To work with these systems, it is essential that you understand how the notation works.

In this notation, simple logical statements such as 'it will rain today' are assigned variables and are connected by logical operators, just like how variables and mathematical operators are used in maths to construct algebraic expressions and equations like *2p/q* or *4x + 5y = 7*.

Here, you are given a list of weather forecasts in English, followed by their 'translations' in the logical notation in **random order**. Variables are arbitrarily assigned to logical statements. Keep in mind that *or* is inclusive throughout this exercise (i.e. x *or* y means x *or* y *or both*).

1. It will rain today but not tomorrow.
2. If it rains today or there is both cold weather today and no snow tomorrow, it will be warm tomorrow.
3. If it rains today and tomorrow, it will be cold tomorrow but it will not snow.
4. It will not snow both today and tomorrow (but it might snow one of the days).
5. If it rains tomorrow, either it will not snow tomorrow or it will rain today.
6. It will be warm today or it will be cold and either snow or rain tomorrow.
7. There will be no snow today or tomorrow.
8. It is either warm today or it will neither rain today nor tomorrow.
9. If it rains or snows today, there will be cold weather tomorrow.
10. It will not be cold today.
11. It will be cold today and if it doesn't rain today, it will snow tomorrow.

(a) `AsNApw` (b) `AsKqAtw` (c) `CApuq` (d) `CApKvNtr` (e) `Nv` (f) `CKpwKqNt`
(g) `NKut` (h) `KvCNpt` (i) `KpNw` (j) `CwANtp` (k) `NAut`

Assignment 1:

Identify the correspondences between the forecasts (1-11) and their representations in logic notation (a-k).

[1] Polish notation (PN), also known as normal Polish notation (NPN), Łukasiewicz notation, Warsaw notation, Polish prefix notation or simply prefix notation, is a mathematical notation in which operators precede their operands. When Polish notation is used as a syntax for mathematical expressions by programming language interpreters, it is readily parsed into abstract syntax trees. Because of this, Lisp and related programming languages define their entire syntax in prefix notation.

Assignment 2:

Convert these statements in logic notation to weather forecasts in English:

a. `KNpq`

b. `CAuNrw`

c. `KNsCNpw`

Assignment 3:

Convert these weather forecasts into logic notation.

(a) Either it will be neither cold nor warm tomorrow, or it will rain tomorrow.

(b) It will snow today and if it is cold tomorrow, it will snow tomorrow.

(c) If it rains today but not tomorrow, it will not be cold either today or tomorrow (but it could be cold one of the days).

Assignment 4:

Explain briefly, in your own words, what the following symbols are used for in the logical notation:

(1) `A` (2) `C` (3) `K` (4) `N`

– Ishan Ganguly

Note: Since PLO 2021 Round 1 was held as an online contest, this problem was presented in an obfuscated form, with the real facts about the language modified/omitted.

Kholosi

The following is a scrambled list of Kholosi[1] sentences and their English translations.

1. ɔnguloɔn
2. hiki sokoli gazzi
3. mɔy hiki motor ke pendemay
4. ho tɔffen ke cʰenemay
5. vyɔw ɔngulos te gazzien ke cʰenai
6. tiriṛi śurkiu tiriṛi motor ke pendavɔn
7. Nilufar segaw tɔffen ke pesaiya
8. tereḷɔ javɔn vyɔwkɔ Ali ke barɔ ketɔb genavɔn
9. mɔy gohroy kɔro e
10. śurki motoren ke pendai
11. Nilufar gohros sokolo e

(a) Nilufar is able to see apples.

(b) I am looking at one car.

(c) My horse is black.

(d) The three young boys gave Ali two books.

(e) The girl saw cars.

(f) one white rope

(g) The boy cut the ropes with his finger.

(h) Nilufar’s horse is white.

(i) He is cutting the apples.

(j) Three girls saw three cars.

(k) their finger

Assignment 1:

Match the Kholosi sentences with their correct English translations.

[1]Kholosi is an Indo-Aryan language spoken in two villages (Kholus and Gotav) in Hormozgan, Iran.

Assignment 2:

Translate into Kholosi.

(a) their horse

(b) Three girls cut the ropes.

(c) Three boys cut the ropes.

(d) Ali’s apple is white.

(e) He cut the apples.

(f) The girl saw the rope.

(g) The girls saw the apples.

(h) The girls gave the rope to Nilufar.

– Aryaman Arora

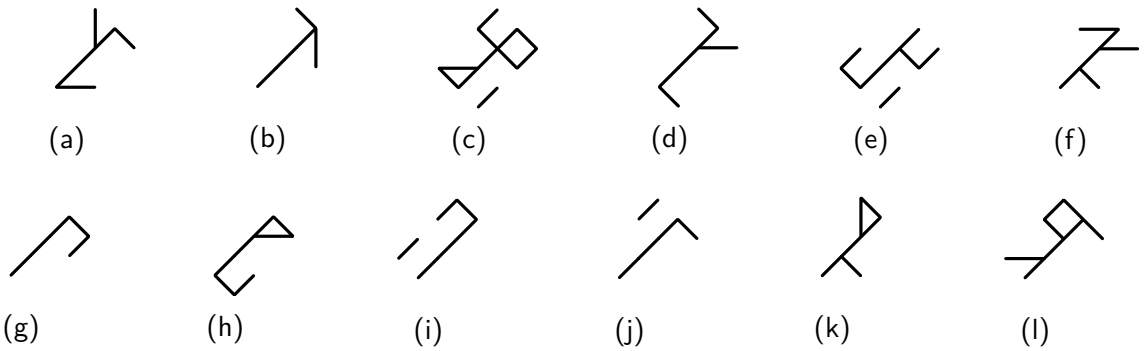
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Cistercian Numerals

A long time ago, hundreds of monks lived in exile, and needed to communicate with their friends on the outside. They had secret channels passing through trade routes, but there was always the risk of interception. To protect their information, they encoded their messages using a cipher.

You are a friend of the community, and new to the practice of trade, so you do not properly know the cipher. You are carrying a message that needs to go across a trade route. Unfortunately, you believe there will be guards at the next crossroads, and you don't want to be caught carrying secrets. Before crossing, you must decipher the message and memorize it in your head, destroying all evidence of such communication.

Luckily, you have mostly figured out the cipher, except for the numbers. Below are some numbers that appear in the message written using the ✠ Cistercian numerals ✠ notation[1]. (The numbers are written in no particular order.)



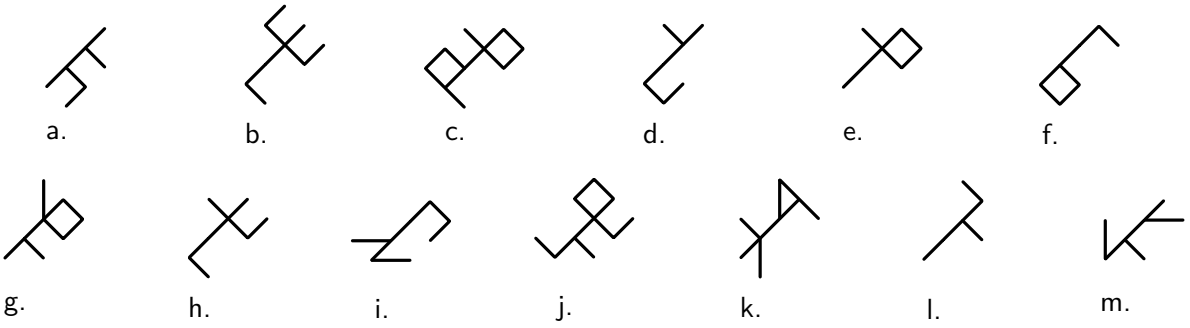
You are confident that their counterparts are the following numbers (written in ascending order):

7, 13, 61, 114, 234, 250, 341, 705, 4091, 5689, 6070, 7608

Assignment 1: Your task is to match the ✠ Cistercian numerals ✠ with their counterparts.

[1] The medieval Cistercian numerals, or "ciphers" in nineteenth-century parlance, were developed by the Cistercian monastic order in the early thirteenth century at about the time that Arabic numerals were introduced to northwestern Europe.

Assignment 2: Now back from your journey, you must relay a message from the outside to the community in exile. How would you write down today's date (249/12/1298) on the message in the ✠ Cistercian numerals ✠ notation? (There is a typo in the date, but encode it as-is, regardless). Pick three numbers, one each for a date, a month, and a year, from the following collection and indicate which is which (e.g., date: x., month: y., year: z.).



– Aalok Sathe

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Kubeo

Given below is a list of Kubeo[1] words with their translations, in **no particular order**. Note that two Kubeo words given translate to 'cave' in English.

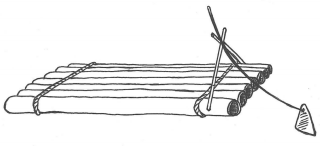
You are given the fact that jabe means 'seed' in Kubeo. You are also given some pictures that illustrate what some of these words mean. Using this information, match the words in the lists given below.

(a) dẽĩjabe (b) dẽĩkũ (c) xiajokũ (d) tãũjo (e) tãũbɨ (f) hokebɨ (g) ũejo (h) ũekobe
(i) kɨ̃ɨ̃kobe (j) kɨɨ̃jãbĩ (k) kɨɨ̃bo (l) jãbĩ (m) xia

(1) river (2) house (3) cave (4) cave (5) rat's nose (6) heap of moriche fruit (7) fishtrap
(8) nostril (9) raft (10) iron nail (11) rock (12) metal can (13) moriche seed

Note: Moriche palm is a palm tree species that grows in and near swamps and other wet areas in tropical South America.

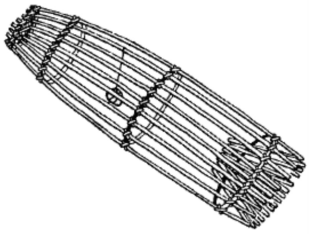
A raft is a means of water transport which has traditionally been constructed of wood or reeds tied together to stay afloat a water body.



Raft



Rat



Fishtrap

Assignment 1:

Here is a list of some of the units of meaning that make up these words. Guess what they mean on their own.

(1) dẽĩ (2) kũ (3) jo (4) tãũ (5) bɨ (6) ũe (7) hoke (8) kobe (9) kɨrɜ̃

[1]Kubeo is a language from the Eastern Tukanoan family, spoken in the Northwest Amazon.

Assignment 2:

Here are the meanings of some additional words in Kubeo:

Kubeo	English
toiwai	writing
bẽ	thread
bú͡tʃi	tobacco
hɨoi	healing
jukira	salt
toa	fire, light

Using these, translate the following Diaboa words to English:

(1) toiwaixũe (2) tãũbẽ (3) hɨoijãbĩ (4) jukirajabe (5) toabɨ

– Abhinav Menon and Mukund Choudhary

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Esimbi

Here are two sequences of five numerals written in Esimbi[1]. In each sequence, the ratio between each number and the next is the same. Note that this ratio may or may not be different across the two sequences.

Sequence 1

(a) mərakpə

(b) moŋi

(c) moŋioŋi

(d) buɣu nə malalə

(e) mobuɣu makələ nə mərakpə

Sequence 2

(a) kenə

(b) makələ

(c) matənə moŋi

(d) mobuɣu mɔrakpə nə matənə mərakpə

(e) mobuɣu moŋioŋi nə kenə

Assignment 1:

Find the Hindu-Arabic numerals (i.e. the numbers you usually write) that correspond to the two sequences above.

Assignment 2:

Following are three expressions in Esimbi, where the blank boxes are placeholders for the missing operators. All three expressions evaluate to the same number, 42.

(a) mobuɣu buɣu nə mobuɣu mɔrakpə nə malalə ☐ makələ

(b) mobuɣu matənə nə kenə ☐ matənə moŋi

(c) mobuɣu mɔrakpə nə kenə ☐ mərakpə

Write 42 in Esimbi numerals. Fill in the missing operators and write the equations in Hindu-Arabic numerals.

– Mukund Choudhary and Aparna Gupte

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[1]Esimbi is a Tivoid language of southwestern Cameroon. It is also called Isimbi, Simpi, Age, Aage, Bogue and Mburugam.